

EE 3201.302

Lab 11 RLC Steady State Analysis

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I. Abstract

This laboratory focuses on the steady-state analysis and characterization of RL, RC, and RLC circuits operating under sinusoidal conditions, with an emphasis on their behavior as filters—specifically low-pass, high-pass, band-pass, and band-stop configurations. The study integrates theoretical modeling through PSpice simulations to predict the frequency-to-voltage response of each filter. These simulations are experimentally validated using physical components, a network analyzer, and Analog Discovery 2 (AD2) with Waveforms software to measure and compare phase and output voltages. By analyzing and comparing simulated and measured results, students gain insight into practical circuit behavior and model accuracy. The principles explored have broad applications in areas such as impedance matching, and tuning circuits for radio, television, and general signal processing systems.

II. Introduction

This laboratory focuses on the sinusoidal steady-state analysis of RLC circuits composed of resistors (R), inductors (L), and capacitors (C) to examine their filtering characteristics. Specifically, it investigates the behavior and performance of four fundamental filter types: low-pass, high-pass, band-pass, and band-stop. While DC analysis applies to systems powered by constant voltages or currents, and transient analysis applies to systems with abrupt changes, this lab emphasizes steady-state conditions where input sources vary sinusoidally, and all transient effects have dissipated.

Under such conditions, the voltage across and current through each component oscillate at the same frequency as the source, though their amplitude and phase vary depending on the configuration and component values. Through PSpice simulations and physical circuit construction, students will analyze how amplitude and phase respond across a frequency range from 1 Hz to 10 MHz. Using the Analog Discovery 2 and waveforms software, experimental measurements will be compared against simulated results to validate the circuit behavior. The lab also includes determining the cutoff frequency, defined as the point where the output amplitude drops to 0.707 of its maximum, which is a key parameter in filter performance analysis.

III. Procedure

1. Set up AD2 Waveforms Software
 - a. Use the Network Analyzer in Waveforms; set scale to logarithmic, frequency range from 100 Hz to 2 MHz, and samples to 151.
 - b. In Wavegen, set offset to 0 V, mode to constant, and amplitude to 1 V.
 - c. Set magnitude units to gain (X), top to 2 X, bottom to 0 X, and select “Relative to Channel 1.”
 - d. Set phase offset to 0°, range to 360°; turn off Channel 1 and set Channel 2 offset to 0 V and gain to 1 X.

2. Build Circuit and Collect Data

- For each the RL filters, RC filters, and RLC bandpass filters shown below, build based off the schematics
- Capture the data to view in the Networks tab of Waveforms
- Use the cursors to mark the cutoff resonant frequencies of the output
- Record the results (either through exporting as a screenshot, or as pure data)
- In the Networks tab, set the stop frequency to be 5 MHz, and i.e. Steps to 501
- Build the circuit shown in Figure 3.8
- Follow through steps 2-4, making sure to also measure the resonant frequency
- Now, in the Networks tab, set the number of samples (i.e. Steps) to 151
- For each circuit in Figure 3.6 and Figure 3.7, construct according to the schematic
- Follow through steps 2-4, making sure to also measure the resonant frequency

1. Low Pass Filters

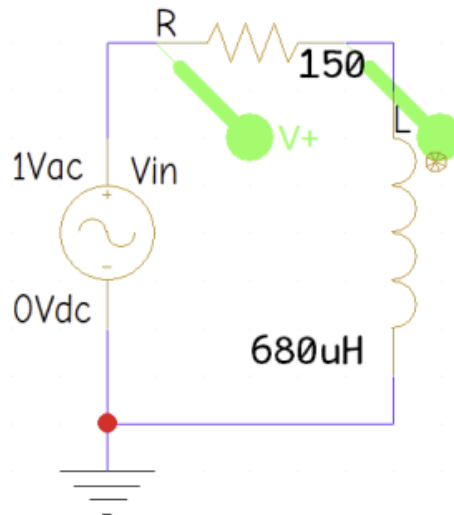


Figure 3.1: RL Low Pass Filter Schematic

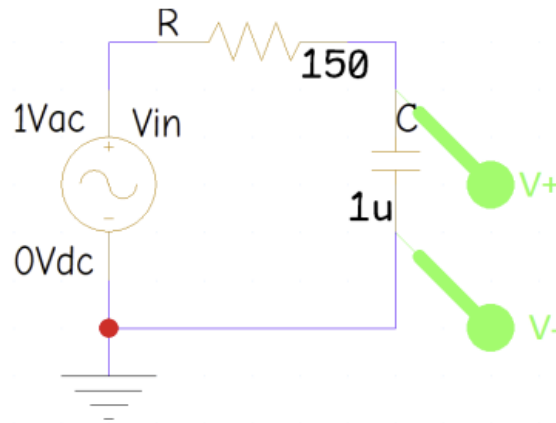


Figure 3.2: RC Low Pass Filter Schematic

2. High Pass Filter

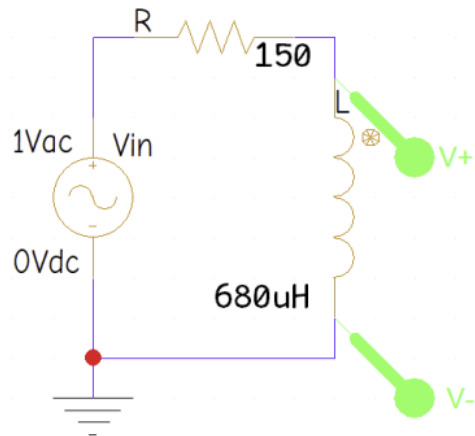


Figure 3.3: RL High Pass Filter Schematic

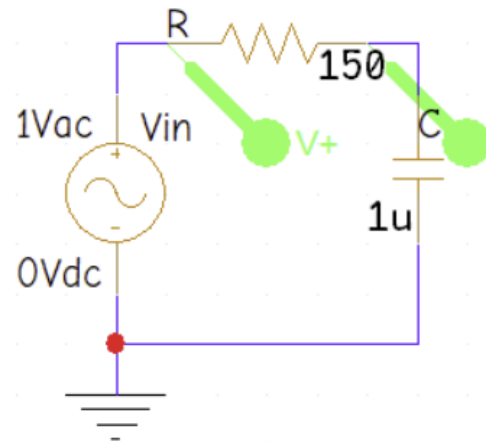


Figure 3.4: RC High Pass Filter Schematic

3. Band Pass Filter

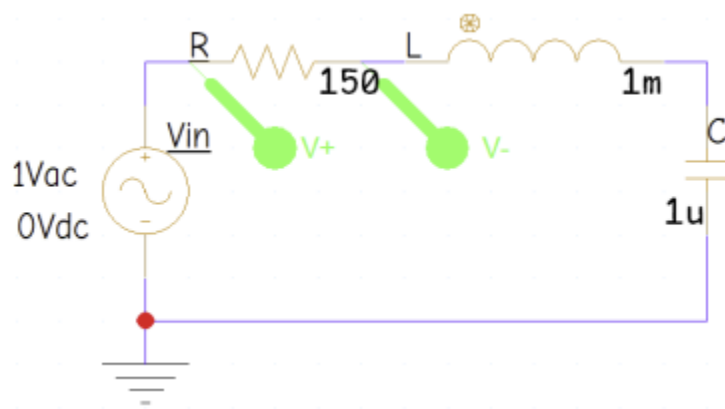


Figure 3.5: RLC Band Pass Filter Schematic

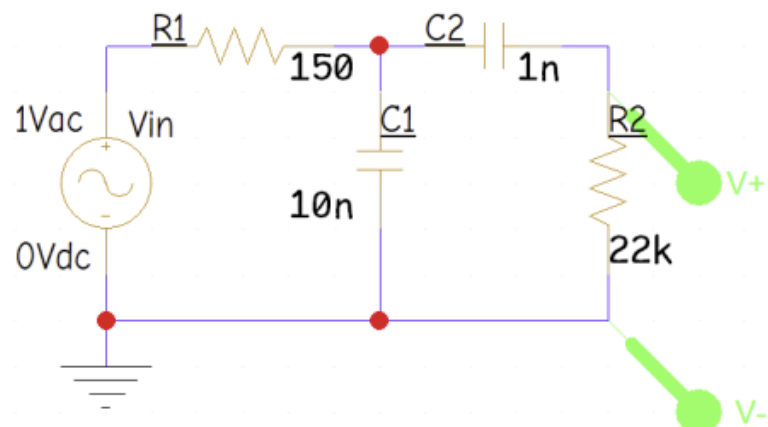


Figure 3.6: RC Band Pass Filter (low to high) Schematic

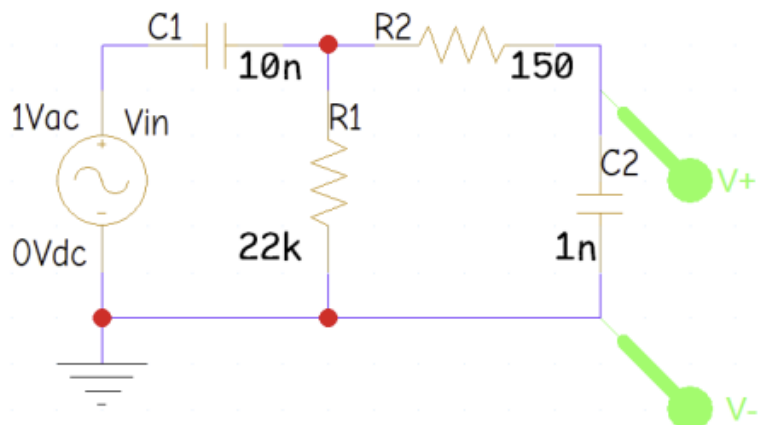


Figure 3.7: RC Band Pass Filter (high to low) Schematic

4. Band Stop Filter

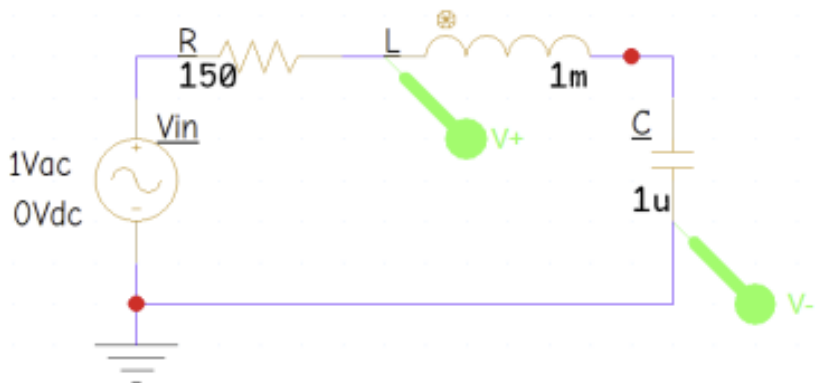


Figure 3.8: RLC Band Stop Schematic

IV. Experimental Results

1. Low Pass Filter

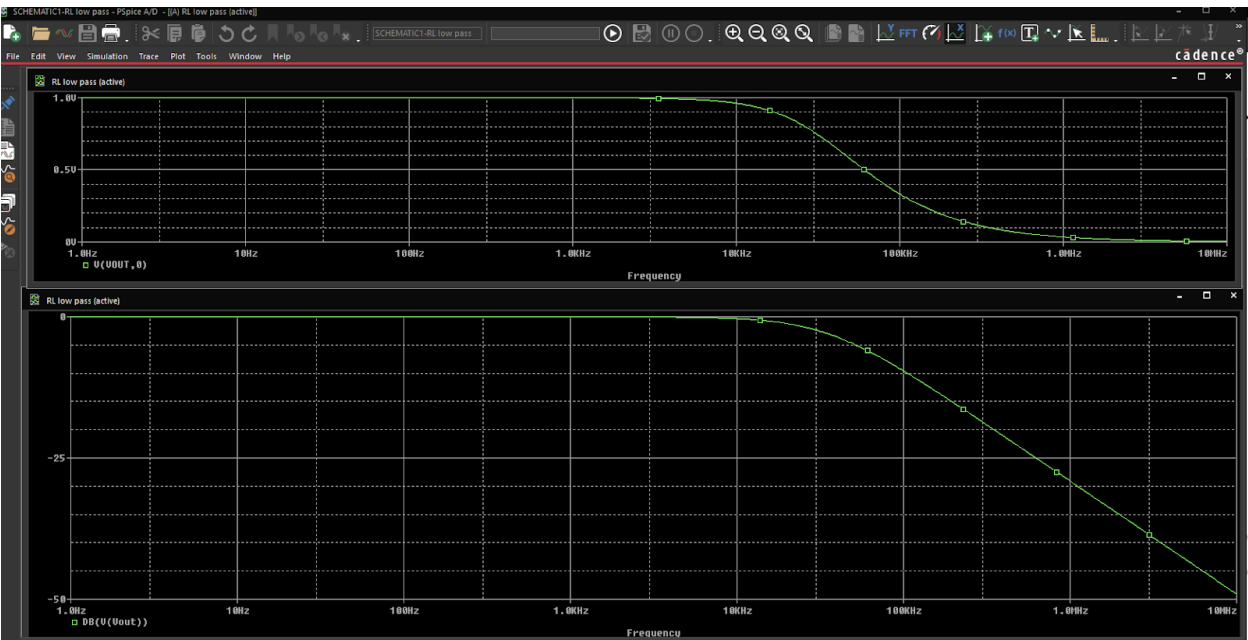


Figure 4.1: RL Low Pass Filter PSpice Simulation

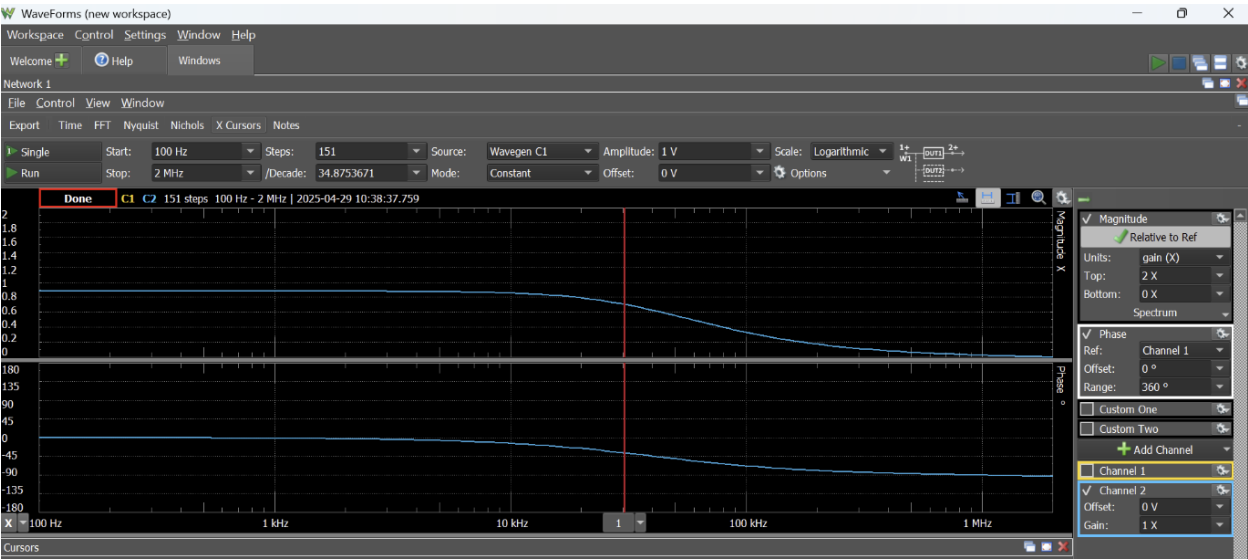


Figure 4.2: RL Low Pass Filter Results

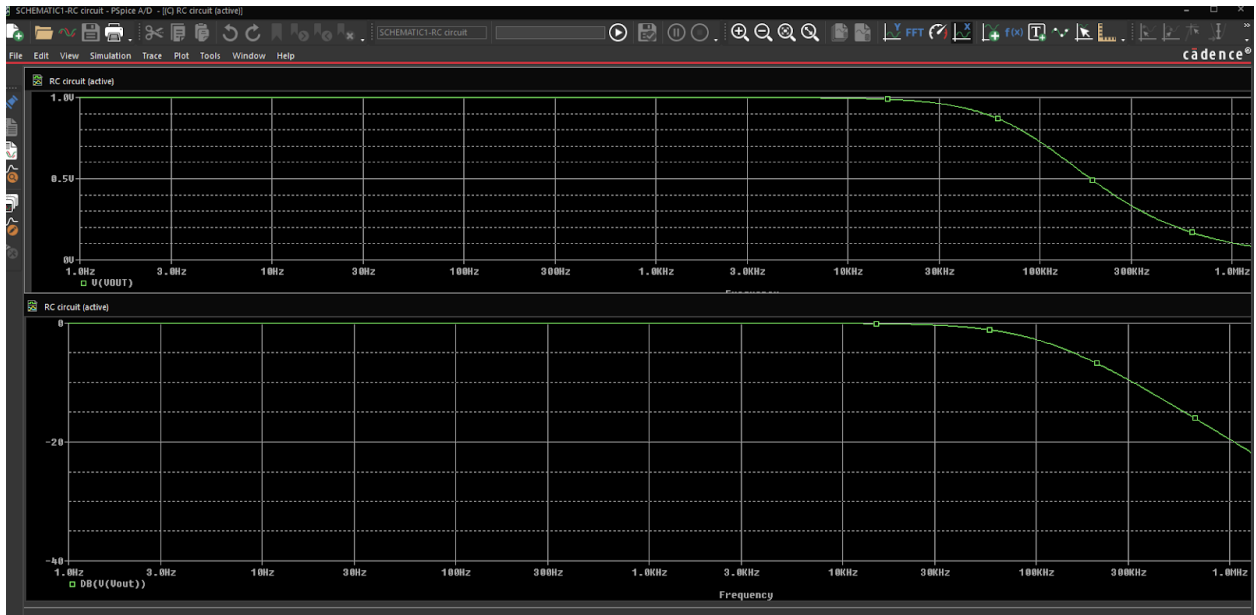


Figure 4.3: RC Low Pass Filter PSpice Simulation

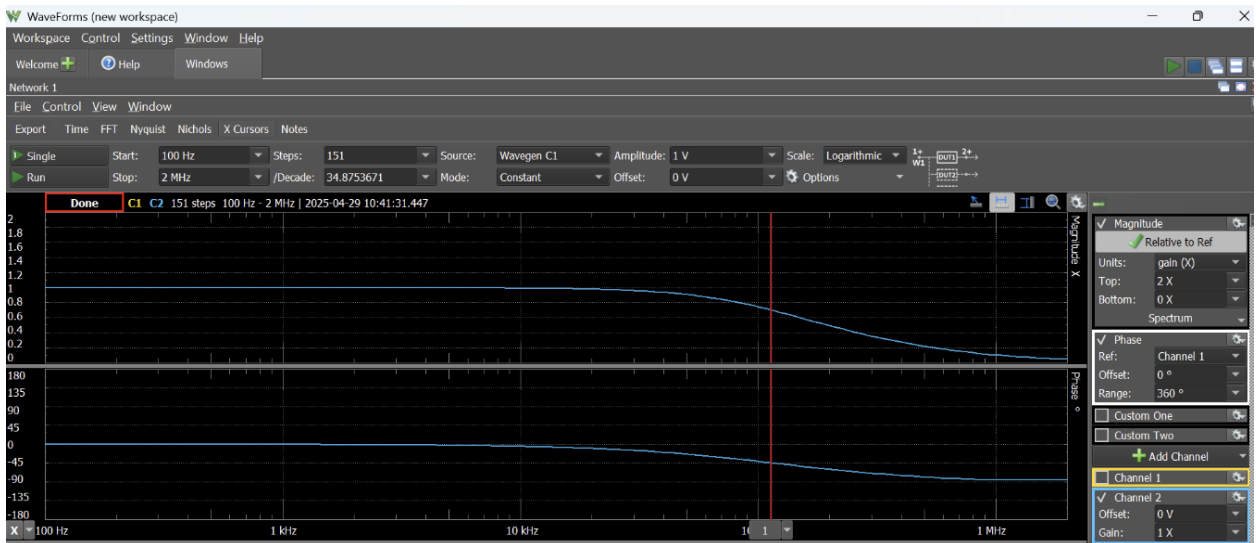


Figure 4.4: RC Low Pass Filter Results

2. High Pass Filter

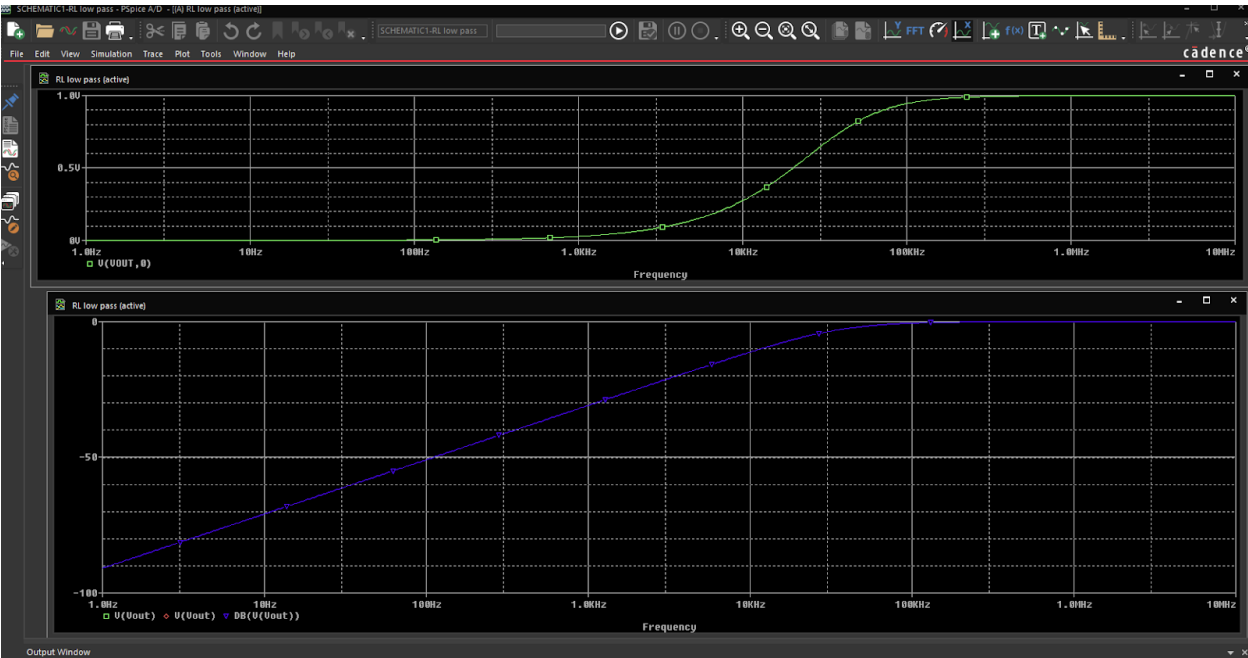


Figure 4.5: RL High Pass Filter PSPice Simulation

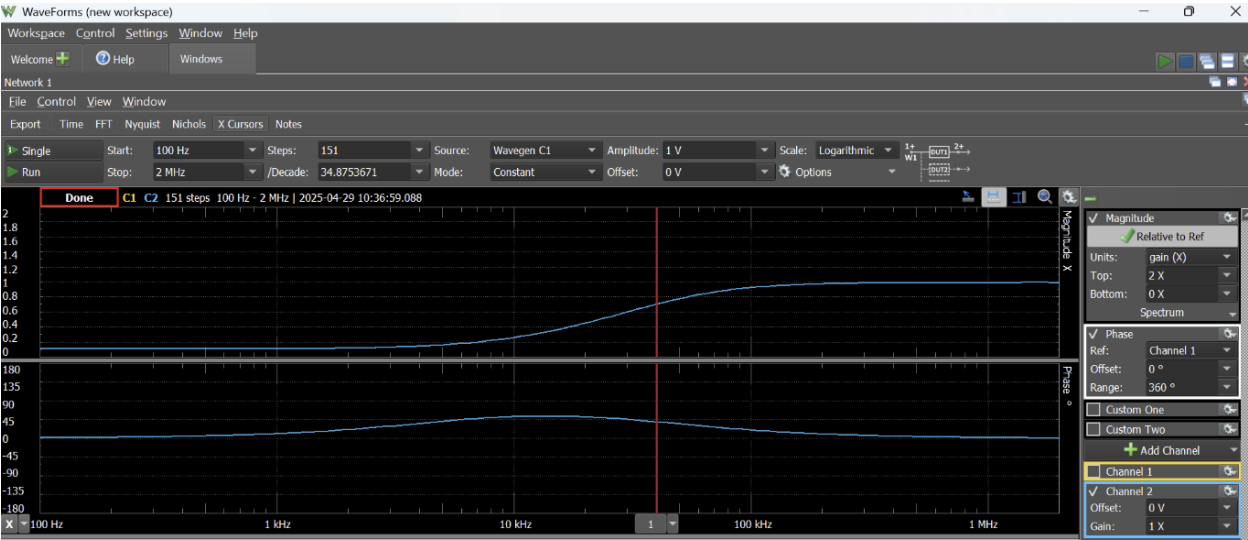


Figure 4.6: RL High Pass Filter Results

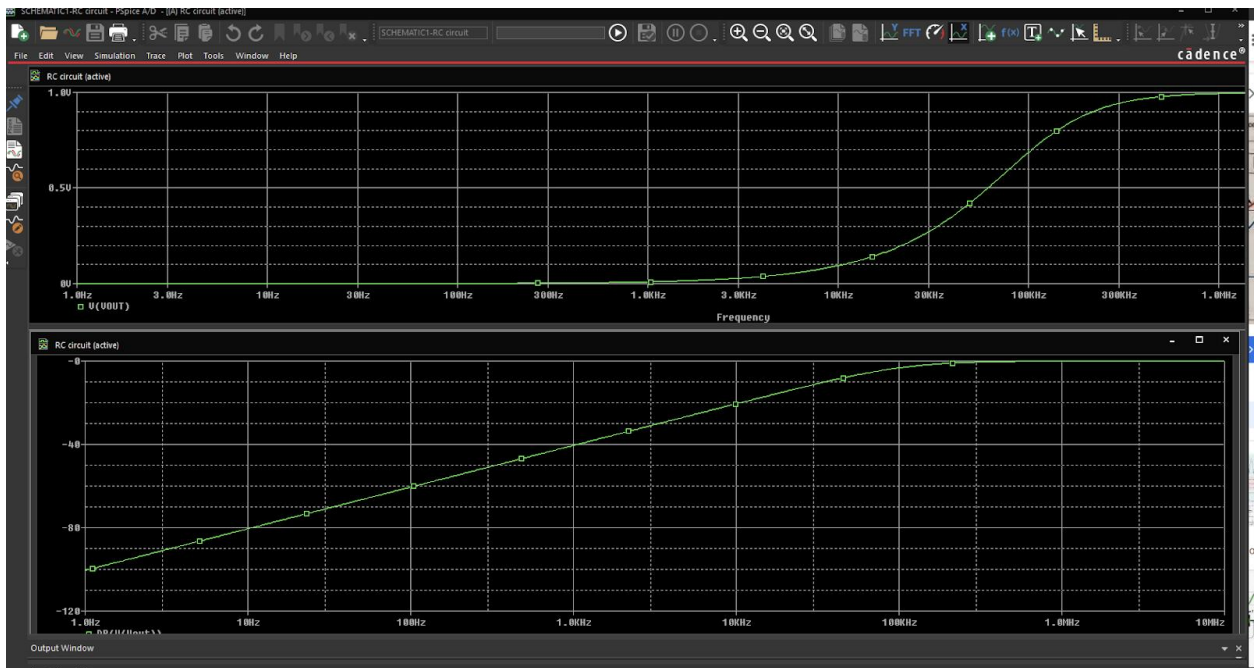


Figure 4.7: RC High Pass Filter PSPice Simulation

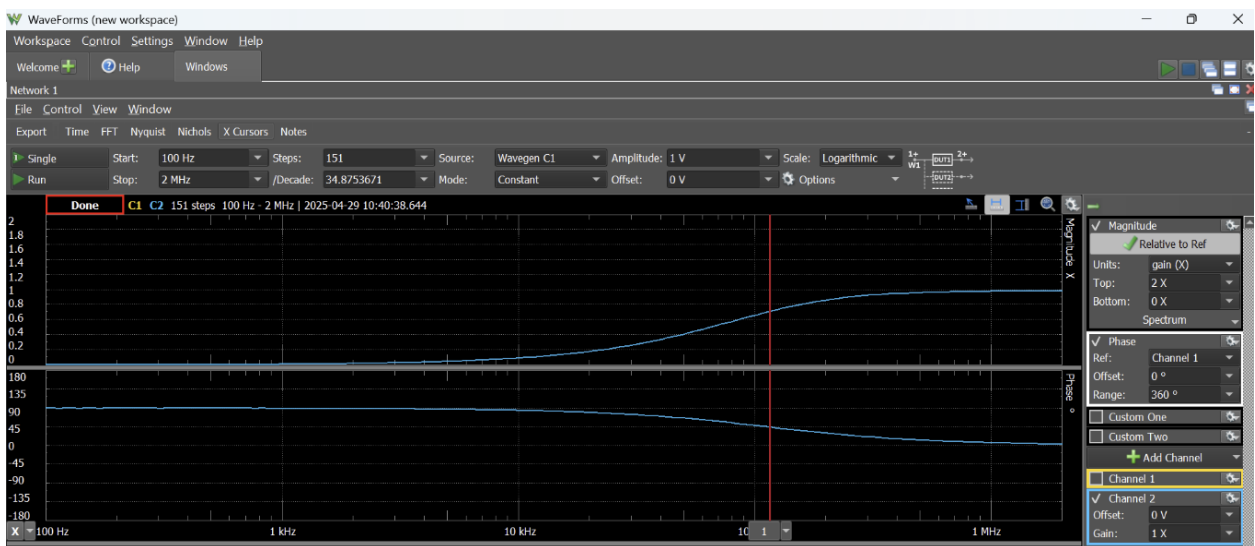


Figure 4.8: RC High Pass Filter Results

3. Band Pass Filter

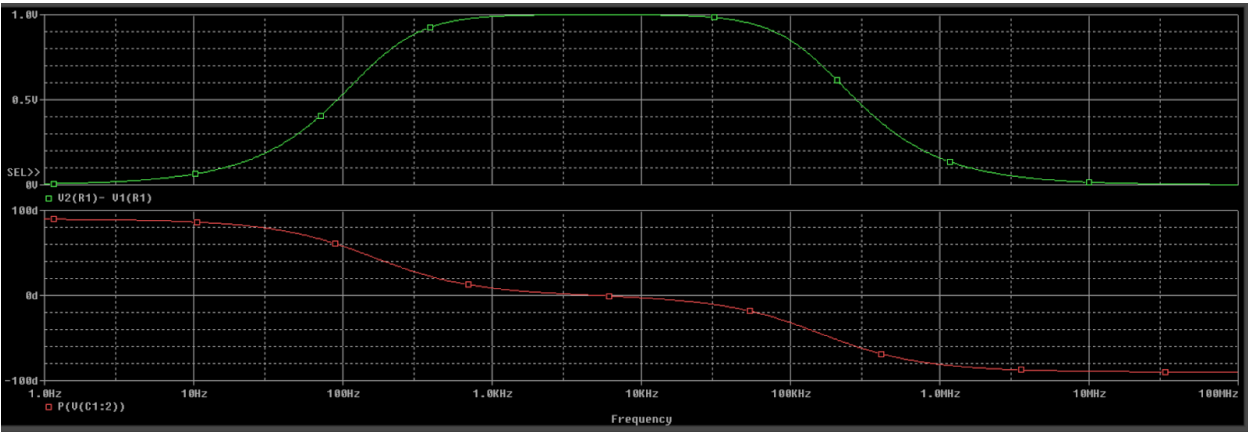


Figure 4.9: RLC Band Pass Filter PSpice Simulation

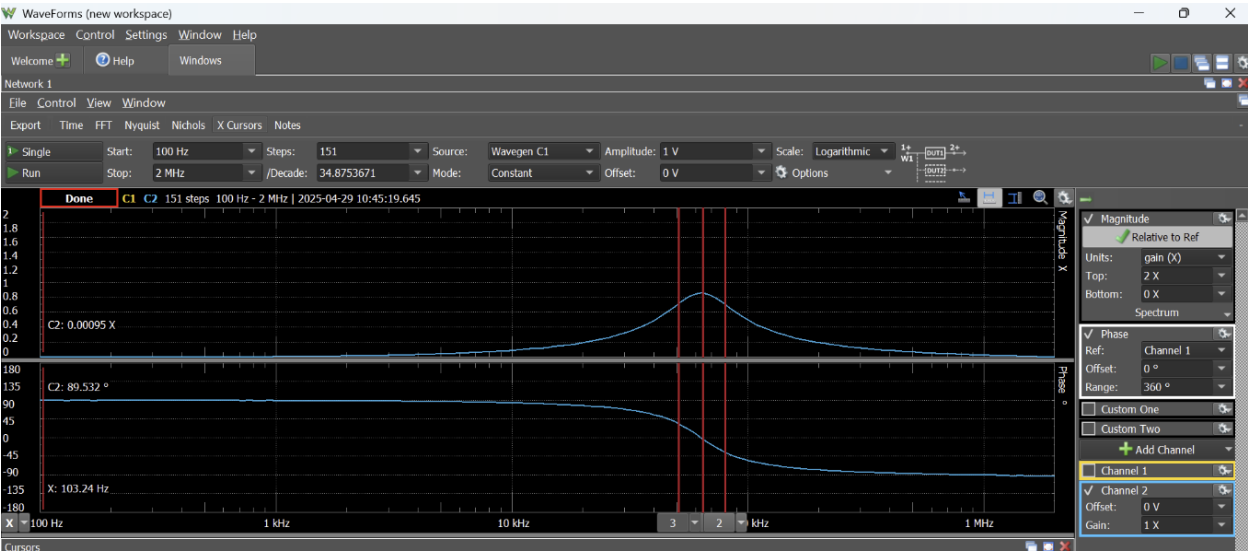


Figure 4.10: RLC Band Pass Filter Results

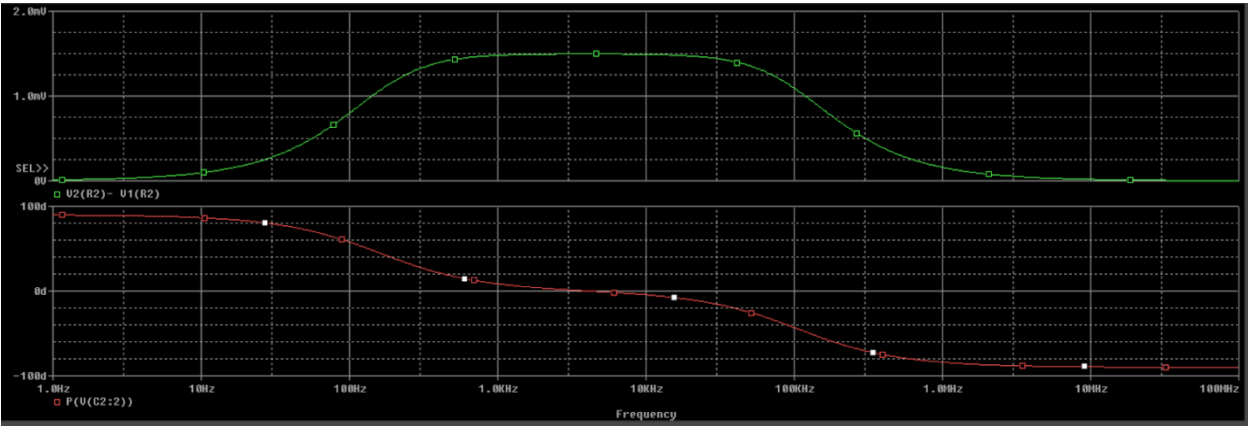


Figure 4.11: RC Circuit as a Band Pass Filter (low to high) PSpice Simulation

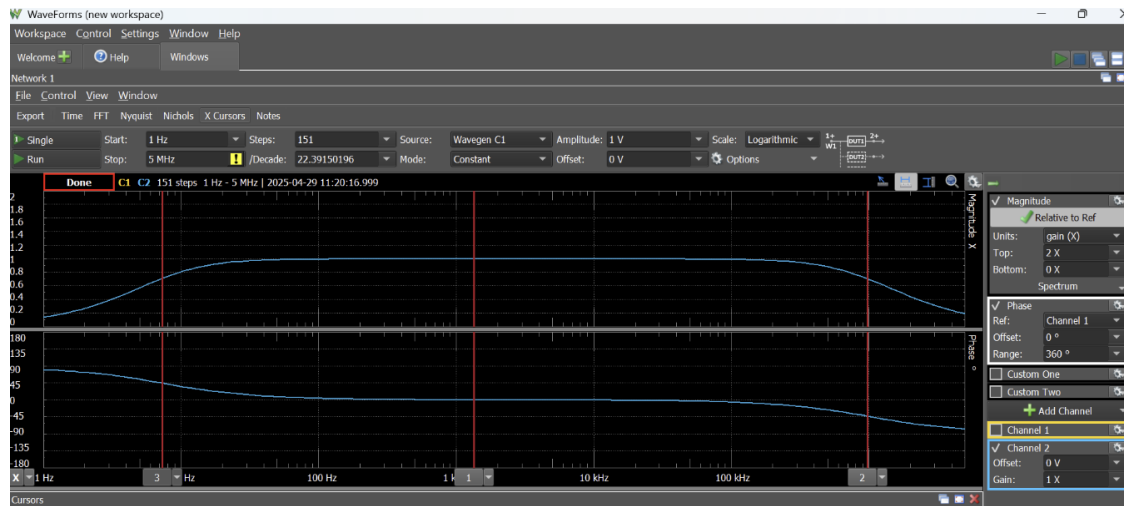


Figure 4.12: RC Circuit as a Band Pass Filter (low to high) Results

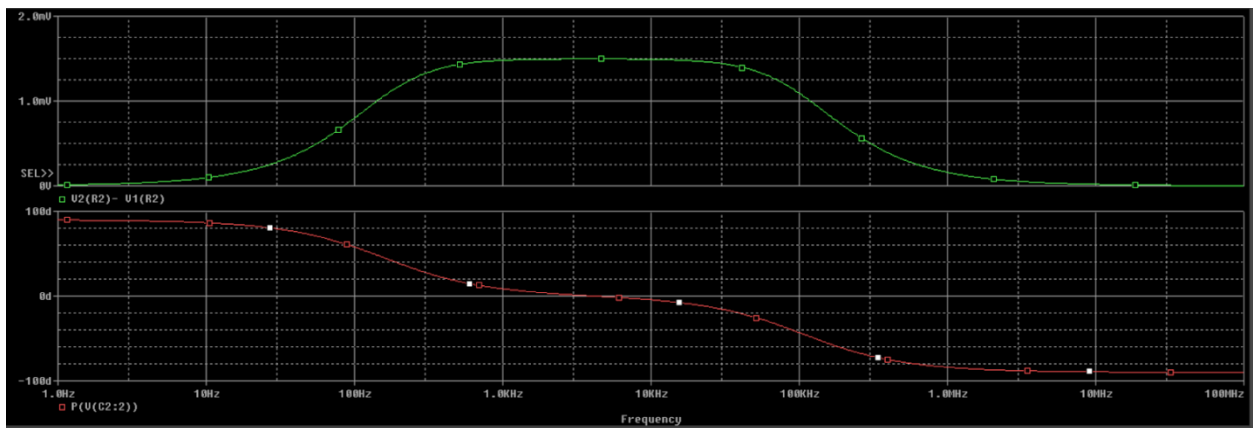


Figure 4.13: RC Circuit as a Band Pass Filter (high to low) PSpice Simulation

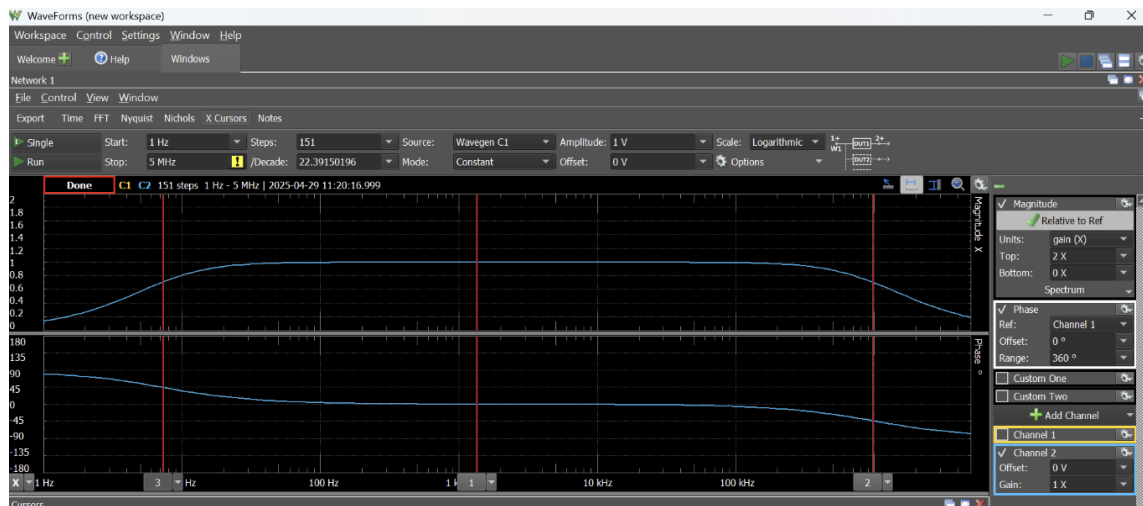


Figure 4.14: RC Circuit as a Band Pass Filter (high to low) Results

4. Band Stop Filter

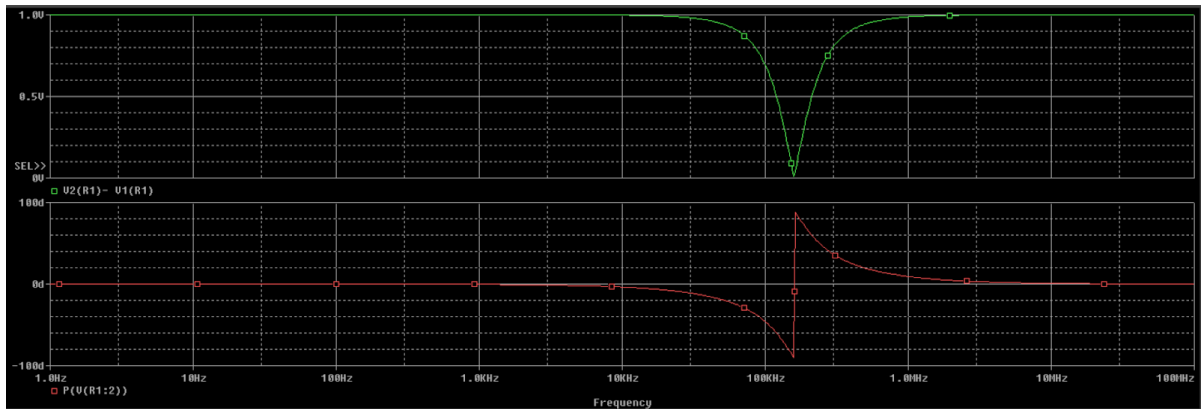


Figure 4.15: RLC Bandstop Circuit PSpice Simulation

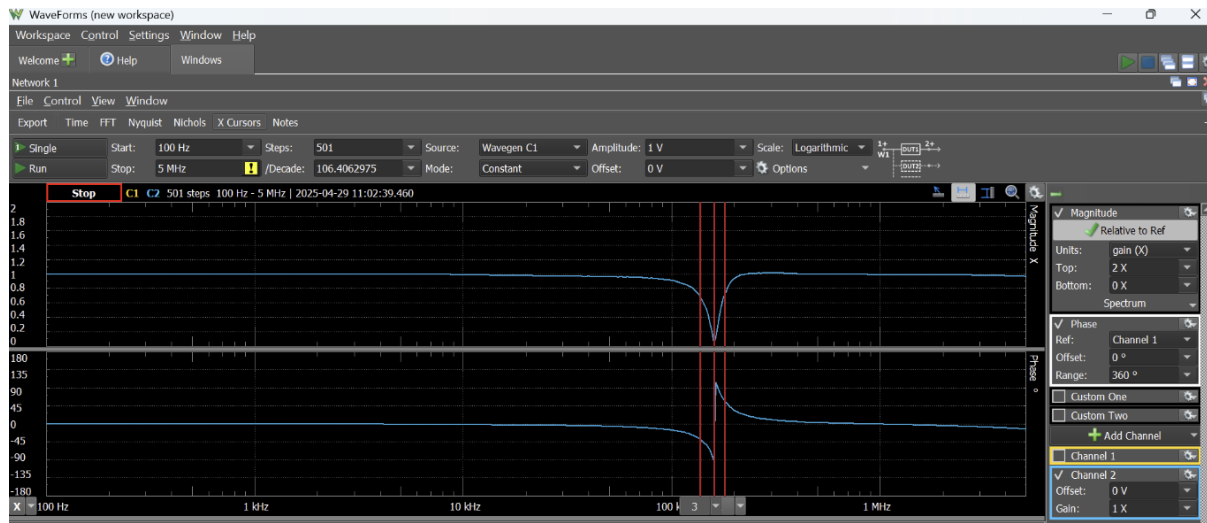


Figure 4.16: RLC Bandstop Circuit Results

V. Thoughtful Questions

1. The phase shift and output amplitude (gain) you obtained in lab differ from those shown in your PSpice simulations. This is a common occurrence when designing RLC circuits in industry. What are some possible reasons why this could be the case?

The phase shift and gain measured in the laboratory may differ from the PSpice simulations due to real-world component tolerances actual component values often vary slightly from their nominal ratings. Additional discrepancies can arise from noise measurement, as instruments like the AD2 introduce some input impedance that can affect readings. User error is also a factor, particularly when identifying cutoff and resonant frequencies. For example, the oscilloscope screenshots reveal that the measured cutoff points often do not correspond exactly to when the gain drops to 0.707 of the peak value.

2. Were the results from your RC band-pass circuits (the last two circuits) similar? Why was this the case?

The results obtained from the RC band-pass circuits were consistent across both configurations, as identical component values were used in each case. Since the cutoff and resonant frequencies are determined by these component values, the frequency response remains effectively unchanged regardless of the filter order (i.e., high-pass followed by low-pass or vice versa). Although the arrangement may introduce slight variations in the phase relationship of signals within the passband, the overall amplitude response and cutoff frequencies remain the same. This demonstrates that in linear time-invariant systems, the ordering of cascaded filters with the same frequency characteristics does not significantly affect the magnitude response.

VI. Discussion and Conclusions

The results of this laboratory confirm the theoretical behavior of RL, RC, and RLC circuits under sinusoidal steady-state conditions. By analyzing various filter types low-pass, high-pass, band-pass, and band-stop we observed how amplitude and phase shift vary across frequency, and how these variations are influenced by the type and configuration of circuit components.

One key observation was the general agreement between PSpice simulations and experimental measurements, particularly in identifying cutoff and resonant frequencies. However, some discrepancies were present. These differences are attributable to component tolerances, which cause real-world components to deviate slightly from their nominal values. Additionally, Analog Discovery 2 introduces noise measurement and input impedance that can affect reading. User error, especially when identifying cutoff or resonant frequencies using oscilloscope cursors, also contributed to variation. For instance, the measured gain at the cutoff point was not always exactly 0.707 of the peak value, which slightly affected accuracy.

The RC band-pass circuits (configured as low-pass followed by high-pass and vice versa) provided nearly identical results. This validates that in linear time-invariant systems, the order of cascading filters does not significantly influence the overall magnitude response, though small differences in phase may occur. Since the same component values were used in both configurations, their frequency characteristics—such as cutoff and resonant points—remained consistent.

Overall, this lab reinforced the practical understanding of sinusoidal steady-state analysis and demonstrated the usefulness of simulation tools in circuit design. It also highlighted the importance of accounting for real-world factors—such as tolerance, noise, and measurement techniques when analyzing and interpreting experimental data. These insights are essential for designing reliable filters in real-world applications like signal processing, impedance matching, and communications systems.